The Word-Complexity Ladder for Collatz Divergence: a Machine-Verified Rung Zero and a Mahler-Value Reduction of the Automatic Rung

M. Sharpe*

June 12, 2026

Abstract

Order potential divergent orbits of the Terras map ($T(n) = n/2$ for even n, $(3n+1)/2$ for odd n) by the complexity of their parity itineraries. We prove the floor of this ladder formally: a divergent orbit's itinerary is never eventually periodic (machine-verified in Lean 4, axiom-free). We then reduce the next rung to a single arithmetical statement. For the fixed point w of the substitution $1 \mapsto 110$, $0 \mapsto 011$ — an aperiodic automatic word of parity density $2/3$, inside the supercritical band where divergent orbits must live — the generating function $F(z) = \sum w_i z^i$ satisfies the Mahler functional equation $F(z) = (z+z^2)/(1-z^3) + (1-z^2)\,F(z^3)$, and the unique 2-adic integer whose itinerary is w equals $-10 + \frac{5}{9}F_2(8/9)$, where F_2 denotes 2-adic evaluation. Consequently, if the 2-adic Mahler value $F_2(8/9)$ is irrational, no integer has itinerary w. The archimedean value $F(8/9) \in \mathbb{R}$ is transcendental by Mahler's classical method; the 2-adic value is the open case, and we prove that it is genuinely beyond naive methods: the truncation approximations to $F_2(8/9)$ have irrationality exponent exactly $(3\log 2)/(2\log 3) = 0.9464 < 1$, so no Liouville-style argument can decide the question — and the deficiency constant is, once again, the critical density of the $3x+1$ problem. The same reduction applies uniformly to every constant-length substitution with constant block weight in the supercritical band, where double (real and 2-adic) convergence of the relevant series is automatic.

1 Introduction

The Collatz conjecture asserts that iterating the Terras map

$$T(n) \;=\; \begin{cases} n/2 & n \text{ even}, \\ (3n+1)/2 & n \text{ odd}, \end{cases}$$

from any positive integer eventually reaches 1 [6]. Its negation has two faces: a nontrivial cycle, or a divergent orbit. This paper concerns the divergent face, through the lens of symbolic dynamics. The *itinerary* of n is the infinite binary word

$$w(n) \;=\; \big(T^i(n) \bmod 2\big)_{i \geq 0},$$

*msharpe@irendi.com. The mathematics, the Lean formalization, and this text were produced in collaboration with Claude (Anthropic). The Collatz conjecture remains open, and nothing in this paper claims otherwise.

the diary of parity choices the orbit makes. Two facts, both formal in the companion repository [10], frame everything below. First, the itinerary determines the integer completely: distinct positive integers have distinct (in fact everywhere eventually differing) itineraries — the infinite-precision form of Terras' parity bijection [11]. Second, a divergent orbit cannot have an arbitrary itinerary: its odd-step density is pinned, in the liminf, to at least the critical value

$$\delta_c \;=\; \frac{\log 2}{\log 3} \;=\; 0.630929\ldots$$

(the density floor of [10]; intuitively, fewer odd steps than that and the halvings win).

The ladder. Order itineraries by word-theoretic complexity: eventually periodic words; automatic words (fixed points of constant-length substitutions and their codings); morphic and Sturmian words; and beyond. Excluding each class from the divergent repertoire is a theorem target. The conjecture — on its divergence side — is the top of the ladder; each rung is finite, honest work. This paper establishes the floor and gives the second rung a precise, classical shape.

Results.

1. *Rung zero, machine-verified* (Section 3). A divergent orbit's itinerary is not eventually periodic. Formalized in Lean 4 (`divergent_itinerary_aperiodic`, file `Ladder.lean` of [10]), depending only on Lean's standard foundations; no axioms, no `sorry`.

2. *The automatic rung is one Mahler value* (Sections 4–5). For the fixed point w of $\sigma : 1 \mapsto 110,\ 0 \mapsto 011$, the unique 2-adic integer with itinerary w is $x(w) = -10 + \frac{5}{9}F_2(8/9)$, where $F(z) = \sum_i w_i z^i$ satisfies $F(z) = (z+z^2)/(1-z^3) + (1-z^2)F(z^3)$. Hence if $F_2(8/9) \notin \mathbb{Q}$, no integer realizes w. The word w is aperiodic (Lemma 4.1), F is transcendental over $\mathbb{Q}(z)$ (Lemma 4.3), the real value $F(8/9)$ is transcendental by Mahler's classical theorem (Theorem 7.1), and the statement generalizes uniformly to all constant-length, constant-block-weight substitutions in the supercritical band (Proposition 5.2).

3. *The cheap route is provably insufficient* (Section 6). The truncations of the defining series give rational approximations to $F_2(8/9)$ with irrationality exponent exactly $(3\log 2)/(2\log 3) < 1$; rational numbers admit approximations of exponent 1. No argument that uses only the convergence speed of the series can decide irrationality. The deficiency ratio is δ_c again, in disguise, and it caps the naive exponent below 1 throughout the supercritical band — the same coincidence that makes the density theory critical makes the transcendence question non-trivial.

We emphasize the modesty of the target: the automatic rung is a countable, measure-zero family of itineraries, and closing it would leave the conjecture wide open (the companion no-go theorem [10] shows no bounded-memory argument reaches the top of the ladder). Its interest is structural — it is the first rung where the divergence problem meets a developed branch of transcendence theory through an exact interface, with every reduction step proved and most of them machine-checked.

2 Itineraries, the correction term, and realizations

Write T^i for the i-th iterate, $j_T(n)$ for the number of odd values among $n, T(n), \ldots, T^{T-1}(n)$ (the odd-step count), and abbreviate the itinerary prefix $w_{<T} = (w_0, \ldots, w_{T-1})$, $w_i = T^i(n)$ mod

2. The dynamics admits an exact linear form (`terras_exact_form` in [10]): for all n, T,

$$2^T\, T^T(n) \;=\; 3^{j_T} n \;+\; d(w_{<T}), \tag{1}$$

where the correction term d depends only on the parity word, via the closed form (`dcoef_closed`)

$$d(w_{<T}) \;=\; \sum_{\substack{i<T\\ w_i=1}} 2^i\, 3^{\#\{i'>i\,:\,i'<T,\ w_{i'}=1\}}. \tag{2}$$

Both are theorems of the formal development; we use them as given.

Now let w be an *infinite* binary word. Since 3 is invertible in $\mathbb{Z}_2$, (1) reads $n \equiv -3^{-j_T} d(w_{<T})$ (mod 2^T), and from (2),

$$3^{-j_T}\, d(w_{<T}) \;=\; \sum_{\substack{i<T\\ w_i=1}} 2^i\, 3^{-(1+\nu_i)}, \qquad \nu_i := \#\{i' < i : w_{i'} = 1\},$$

exactly (the exponent in (2) is $j_T - \nu_i - 1$). The right-hand side is a partial sum of a series whose i-th term has 2-adic absolute value 2^{-i}; it therefore converges in $\mathbb{Z}_2$.

Definition 2.1. The *realization* of an infinite word w is

$$x(w) \;:=\; -\sum_{i\,:\,w_i=1} 2^i\, 3^{-(1+\nu_i)} \;\in\; \mathbb{Z}_2.$$

Proposition 2.2. *A positive integer n has itinerary w if and only if $x(w) = n$ in $\mathbb{Z}_2$. In particular at most one integer realizes any given word.*

Proof. If n has itinerary w then $n \equiv -3^{-j_T} d(w_{<T})$ (mod 2^T) for every T by (1), and the right-hand sides converge to $x(w)$; hence $n = x(w)$. Conversely if $x(w) = n$ then n agrees with $-3^{-j_T} d(w_{<T})$ modulo 2^T for every T; by the parity bijection (the first T parities of an orbit are determined by the starting point mod 2^T, `parity_of_modEq` in [10], after [11]) and an induction on T, the itinerary of n is w. □

Excluding a word w from the divergent repertoire therefore means proving $x(w) \notin \mathbb{Z}_{\geq 1}$ — a statement about one 2-adic number per word.

3 Rung zero

Theorem 3.1 (machine-verified)**.** *Let n have unbounded orbit: for every B there is t with $T^t(n) > B$. Then the itinerary of n is not eventually periodic.*

Proof sketch. If the itinerary satisfies $w_{s+t} = w_{s+P+t}$ for all $t \geq 0$, then the two tail points $u = T^s(n)$ and $v = T^{s+P}(n)$ have identical itineraries at every time; by the infinite-precision parity bijection (two integers with everywhere-equal itineraries agree modulo 2^k for every k, hence are equal — `eq_of_itinerary_eq`) we get $u = v$. The orbit is then eventually periodic, so it is bounded by the maximum of its first $s + P + 1$ values, contradicting unboundedness. The full proof is `divergent_itinerary_aperiodic` in `Ladder.lean` of [10]; it depends only on `propext`, `Classical.choice`, `Quot.sound`. □

Rung zero is elementary, but it sets the pattern: a complexity class of words is excluded by showing its realizations are never positive integers (here, eventually periodic words realize the rational 2-adic points of the dynamics — e.g. $x((110)^\infty) = -5$ — and the positive-integer ones among them are the bounded orbits).

4 The automatic rung: one functional equation

From here on, w is the fixed point of

$$\sigma : \quad 1 \mapsto 110, \qquad 0 \mapsto 011,$$

starting from $w_0 = 1$: $w = 110\,110\,011\,110\,110\,011\,011\,110\,110 \cdots$. Both images have length 3 and contain exactly two 1s, so w has parity density exactly $2/3 > \delta_c$: a word in the supercritical band, admissible (as far as the density theory knows) as the itinerary of a divergent orbit. The substitution structure is equivalent to the *kernel relations*

$$w_{3i} = w_i, \qquad w_{3i+1} = 1, \qquad w_{3i+2} = 1 - w_i \qquad (i \geq 0), \tag{3}$$

read off from $\sigma(1) = 110$, $\sigma(0) = 011$.

Lemma 4.1. *w is not eventually periodic.*

Proof. Suppose $w_{i+P} = w_i$ for all $i \geq N$, with $P \geq 1$ minimal admissible. If $3 \mid P$, write $P = 3Q$: for $i \geq N$, (3) gives $w_{i+Q} = w_{3(i+Q)} = w_{3i+P} = w_{3i} = w_i$ for all $i \geq N$, so $Q < P$ is also an eventual period; hence we may assume $3 \nmid P$. Consider positions $i \equiv 1 \pmod 3$, $i \geq N$; there $w_i = 1$ by (3). If $P \equiv 1 \pmod 3$ then $i + P \equiv 2 \pmod 3$, so $1 = w_{i+P} = 1 - w_{(i+P-2)/3}$, i.e. $w_{t+(P-1)/3} = 0$ for every $t \geq (N-1)/3$ (writing $i = 3t+1$). Then w is eventually all 0s, contradicting $w_{3i+1} = 1$. If $P \equiv 2 \pmod 3$ then $i + P \equiv 0 \pmod 3$, so $1 = w_{i+P} = w_{(i+P)/3}$, i.e. $w_{t+(P+1)/3} = 1$ for all large t: w is eventually all 1s, contradicting $w_{3i+2} = 1 - w_i = 0$ at the infinitely many i with $w_i = 1$. ☐

Proposition 4.2 (the functional equation)**.** *The generating function $F(z) = \sum_{i \geq 0} w_i z^i \in \mathbb{Z}[[z]]$ satisfies*

$$F(z) \;=\; \frac{z + z^2}{1 - z^3} \;+\; (1 - z^2)\, F(z^3).$$

Proof. Group the sum in blocks of three and apply (3):

$$F(z) = \sum_{i \geq 0} z^{3i}\bigl(w_i + z + (1 - w_i) z^2\bigr) = (z + z^2) \sum_{i \geq 0} z^{3i} + (1 - z^2) \sum_{i \geq 0} w_i z^{3i}.$$ ☐

This is the classical Mahler setup [8, 9]: degree $d = 3$, coefficients in $\mathbb{Q}(z)$, iteration $z \mapsto z^3$. (It is also verified coefficient-wise to degree 900 by the scripts of [10], as are all identities below; see Section 9.)

Lemma 4.3. *F is transcendental over $\mathbb{Q}(z)$.*

Proof. The coefficients of F lie in $\{0, 1\}$. If F were rational, its coefficients would eventually satisfy a fixed linear recurrence with constant coefficients; the sequence of state windows (the D most recent coefficients, for D the recurrence depth) takes values in the finite set $\{0, 1\}^D$, so some window repeats, and forward determinism makes the coefficient sequence eventually periodic — contradicting Lemma 4.1. So F is irrational; by Fatou's theorem [4] (a power series with integer coefficients and radius of convergence 1 is either rational or transcendental over $\mathbb{Q}(z)$; see also [3, Ch. 6]), F is transcendental over $\mathbb{Q}(z)$. ☐

Theorem 4.4 (the evaluation identity)**.** *In $\mathbb{Q}_2$ (where $|8/9|_2 = 1/8 < 1$ makes all series below converge),*

$$x(w) \;=\; -10 \;+\; \tfrac{5}{9}\, F_2(8/9),$$

where $F_2(8/9) := \sum_{i\geq 0} w_i (8/9)^i \in \mathbb{Z}_2$ denotes the 2-adic evaluation.

Proof. Each block of three letters contains exactly two 1s, so ν_{3t+r} (the number of 1s before position $3t+r$) equals $2t$ plus the count inside the current block. By (3), if $w_t = 1$ the block is $(1,1,0)$, with 1s at $r = 0$ ($\nu = 2t$) and $r = 1$ ($\nu = 2t+1$); if $w_t = 0$ the block is $(0,1,1)$, with 1s at $r = 1$ ($\nu = 2t$) and $r = 2$ ($\nu = 2t+1$). The block contributions to $\sum_{w_i=1} 2^i 3^{-(1+\nu_i)}$ are therefore

$$w_t = 1: \quad 2^{3t}3^{-(2t+1)} + 2^{3t+1}3^{-(2t+2)} = \left(\tfrac{8}{9}\right)^t \cdot \tfrac{5}{9}, \qquad w_t = 0: \quad 2^{3t+1}3^{-(2t+1)} + 2^{3t+2}3^{-(2t+2)} = \left(\tfrac{8}{9}\right)^t \cdot \tfrac{10}{9}.$$

Grouping consecutive terms of a convergent series in a complete ultrametric field is harmless, so

$$x(w) = -\sum_{t\geq 0} \left(\tfrac{8}{9}\right)^t \left(\tfrac{10}{9} - \tfrac{5}{9} w_t\right) = -\tfrac{10}{9} \cdot \frac{1}{1 - \frac{8}{9}} + \tfrac{5}{9} \sum_{t\geq 0} w_t \left(\tfrac{8}{9}\right)^t = -10 + \tfrac{5}{9}\, F_2(8/9). \qquad \square$$

Remark 4.5. *Iterating Proposition 4.2 at $z = 8/9$ unrolls Theorem 4.4 into a tower: with $u_k := \beta_k^{-1} F_2(\alpha_k/\beta_k)$, $(\alpha_k, \beta_k) = (8^{3^k}, 9^{3^k})$,*

$$u_k \;=\; \frac{\alpha_k(\alpha_k + \beta_k)}{\beta_k^3 - \alpha_k^3} \;+\; (\beta_k^2 - \alpha_k^2)\, u_{k+1}, \qquad \text{e.g.} \quad u_0 = \tfrac{136}{217} + 17\, u_1,$$

with constants of height $\approx 9^{3^{k+1}}$. The tower view makes the height growth of the method visible; the single-equation view is the one the classical theory consumes.

5 The reduction

Corollary 5.1 (rung 1 for w, reduced)**.** *If $F_2(8/9) \notin \mathbb{Q}$, then no positive integer has itinerary w.*

Proof. If $n \in \mathbb{Z}_{\geq 1}$ had itinerary w, Proposition 2.2 and Theorem 4.4 give $F_2(8/9) = \frac{9}{5}(n+10) \in \mathbb{Q}$. $\square$

The same reduction holds across the natural class.

Proposition 5.2. *Let σ be a substitution on $\{0,1\}$ of constant length L whose two image blocks each contain exactly a ones, with $2^L < 3^a$ (equivalently: the fixed-point density a/L exceeds δ_c), and let w be a fixed point with generating function F. Then $F(z) = A_\sigma(z) + B_\sigma(z) F(z^L)$ with $A_\sigma, B_\sigma \in \mathbb{Q}(z)$ explicitly computable from the two blocks, and there are rationals A, B (computable from the blocks; $B \neq 0$ unless the two blocks contribute identically, in which case $x(w) \in \mathbb{Q}$ and the question is decidable directly) with*

$$x(w) \;=\; A + B \cdot F_2\!\left(\frac{2^L}{3^a}\right),$$

the series converging both 2-adically ($|2^L/3^a|_2 = 2^{-L}$) and in $\mathbb{R}$ ($0 < 2^L/3^a < 1$, by supercriticality). Hence irrationality of the single 2-adic Mahler value $F_2(2^L/3^a)$ excludes all integers from realizing w.

Proof. Identical to Proposition 4.2 and Theorem 4.4: blockwise grouping. Each block of L letters contains exactly a ones, so $\nu_{Lt+r} = at +$ (count inside the block), the type-b block ($b \in \{0,1\}$) contributes $z^t C_b$ with $z = 2^L/3^a$ and $C_b = \sum_{r:\,\sigma(b)_r=1} 2^r 3^{-(1+\#\{r'<r:\,\sigma(b)_{r'}=1\})}$, and $x(w) = -\sum_t z^t(C_0 + (C_1 - C_0)w_t) = -C_0/(1-z) - (C_1 - C_0)F_2(z)$. For the functional equation, group F in L-blocks as before. The double convergence is the displayed absolute-value computation; $3^a > 2^L$ is exactly $z < 1$ in $\mathbb{R}$. □

The supercritical band — the only band where divergent orbits can live — is thus exactly the band where the Mahler series converges at *both* places. We do not currently use the archimedean convergence, but the classical method does (Section 7), and the coincidence deserves to be on the record.

6 The gap: truncation provably cannot decide

Why is this not finished? The series defining $F_2(8/9)$ converges fast, and fast rational approximation is the standard engine of irrationality proofs. The following proposition shows the engine stalls — by a precisely measurable margin.

Proposition 6.1. *For $N \geq 1$ let P_N/Q_N with $P_N = \sum_{i<N} w_i\, 8^i\, 9^{N-1-i} \in \mathbb{Z}$, $Q_N = 9^N$, be the N-term truncation of $F_2(8/9)$. Then:*

1. *(quality) $\left|F_2(8/9) - P_N/Q_N\right|_2 = 2^{-3i_0(N)}$ where $i_0(N) \in \{N, N+1, N+2\}$ is the first index $\geq N$ with $w_{i_0} = 1$; in particular the error is never 0 and*

$$\left|F_2(8/9) - P_N/Q_N\right|_2 \;=\; Q_N^{-\kappa_N}, \qquad \kappa_N \;\to\; \frac{3\log 2}{2\log 3} \;=\; 0.946394\ldots$$

2. *(insufficiency) For every rational $\xi = a/b$ and every rational $P/Q \neq \xi$ with Q odd and $|P| \leq Q$,*

$$\left|\xi - P/Q\right|_2 \;\geq\; \frac{1}{(|a| + |b|)\,Q}.$$

Consequently the approximations of part (1) are consistent with $F_2(8/9) \in \mathbb{Q}$: any family of rational approximations can certify irrationality only with exponent > 1, and the truncations achieve 0.9464*.*

Proof. (1) The error is $\sum_{i\geq N} w_i (8/9)^i$; its terms have strictly increasing 2-adic valuations $3i$, so the sum is nonzero with absolute value that of its first nonzero term, 2^{-3i_0}. The word w contains no three consecutive zeros (every block has a 1 in the middle position, by (3)), so $i_0(N) \leq N+2$. Then $\kappa_N = 3i_0(N)\log 2/(2N\log 3) \to (3\log 2)/(2\log 3)$. (2) $\xi - P/Q = (aQ - bP)/(bQ)$; the numerator is a nonzero integer, so $|aQ - bP|_2 \geq |aQ - bP|^{-1} \geq ((|a| + |b|)Q)^{-1}$, while $|bQ|_2 \leq 1$. □

So the question is genuinely beyond the convergence speed of its own defining series: the gap between 0.9464 and 1 must be crossed by structure, not by speed. Note where the constant comes from. For a general admissible pair (L, a) (Proposition 5.2), the N-term truncation gains $LN\log 2$ bits of 2-adic smallness against a denominator of $aN\log 3$ bits, so the exponent is

$$\kappa \;=\; \frac{L\log 2}{a\log 3} \;=\; \frac{\delta_c}{a/L},$$

the critical density divided by the word's density — and the supercriticality condition $a/L > \delta_c$ is *exactly* the statement $\kappa < 1$, with equality on the critical line. The constant that creates the divergence problem also protects this corner of it: throughout the band where divergent orbits could live, truncation falls short of irrationality by construction.

7 The route: Mahler's method, and one open question

Mahler's method [8, 9] was invented for precisely the situation of Section 6: values of functions satisfying $F(z) = A(z) + B(z)F(z^d)$ at algebraic points, where naive approximation falls short. Its auxiliary-polynomial construction converts the functional equation into approximation families whose exponent grows with the auxiliary degree — crossing any fixed Liouville threshold — and its hypotheses are, for our instance, verified above: F transcendental over $\mathbb{Q}(z)$ (Lemma 4.3), the point $8/9$ algebraic with orbit $(8/9)^{3^k}$ avoiding the singularities of the coefficients (the poles of $(z+z^2)/(1-z^3)$ at cube roots of unity and the zeros of $1-z^2$ at ± 1).

At the archimedean place the classical theorem applies as stated:

Theorem 7.1. *The real number $F(8/9) = \sum_i w_i (8/9)^i \in \mathbb{R}$ is transcendental.*

Proof. Mahler's theorem ([8]; in the form [9, Ch. 1]): if $F \in \mathbb{Q}[[z]]$ converges in $|z| < 1$, satisfies a functional equation of the class above, and is transcendental over $\mathbb{Q}(z)$, then $F(\alpha)$ is transcendental for every algebraic α, $0 < |\alpha| < 1$, whose orbit α^{d^k} avoids the finitely many excluded points (poles and zeros of the coefficient functions). Lemma 4.3 and the singularity check above verify every hypothesis at $\alpha = 8/9$. □

The number rung 1 needs is the *same* function at the *same* rational point, evaluated at the other place where the series converges:

Question 7.2. *Is $F_2(8/9) \in \mathbb{Q}_2$ irrational? More generally: does Mahler's method, transposed to $\mathbb{Q}_2$ — auxiliary polynomials, the non-archimedean Schwarz lemma in place of the maximum principle, heights as before — yield irrationality (or transcendence) of $F_p(\alpha)$ for Mahler functions $F \in \mathbb{Q}[[z]]$ at rational α with $0 < |\alpha|_p < 1$ under the hypotheses of Theorem 7.1?*

We are not aware of a published theorem in exactly this form, though p-adic and function-field variants of Mahler's method exist in several settings (see [9] and its references; [5] develops the method over function fields in positive characteristic, and [2] raises the structural theory at the archimedean place to a complete description of algebraic relations). Locating the statement, or transposing the archimedean proof — the ingredients are place-generic, and non-archimedean analysis is, if anything, kinder to the relevant lemmas — is the concrete work item this paper isolates. We record what is at stake: a positive answer to Question 7.2 excludes, by Corollary 5.1 and Proposition 5.2 (plus a finite-system extension to shifted fixed points, standard in the Mahler literature [2]), the entire uniform-substitution stratum of the automatic rung.

Two further remarks on the interface. First, the celebrated transcendence results for automatic *numbers* [1] do not apply here: $x(w)$ is not an automatic number — its binary digits are not generated by an automaton; only the *series coefficients* are automatic, and the base mixes powers of 2 against powers of 3. The functional equation is the correct interface, not the digit expansion. Second, the obstruction is not the word combinatorics: all of Sections 2–6 are exact and verified; the single missing ingredient is non-archimedean.

8 Calibration: the $5x + 1$ test

Any principle that would exclude divergence must fail to "prove" the same for the $5x+1$ map, where orbits (conjecturally, and overwhelmingly in experiment, e.g. from $n = 7$) really do diverge. The present method passes this filter in the correct way. For $5x + 1$, the identical computation realizes the same word w at $x_5(w) = A' + B'F_2(2^L/5^a)$ — the same generating function at a different rational point ($8/25$ for our σ) — and a 2-adic Mahler argument would conclude the same way: *no integer has that exact automatic itinerary under* $5x + 1$. This is consistent with (indeed expected alongside) real divergence: the observed $5x+1$ escapees have generic-looking itineraries of density near 0.49, nowhere near a substitution fixed point. The method excludes specific symbolic classes; it cannot manufacture global convergence — which is precisely why it can be sound. The $3x + 1$-specific content remains where the companion papers put it: in the density floor that pins divergent orbits into the narrow band where the ordered words live.

9 Scope, and the ladder above

What would closing rung 1 actually buy? Directly: no divergent orbit has an itinerary that is a (shifted) fixed point of a uniform substitution — a countable family, measure zero in every sense. The conjecture would remain open at every higher rung: morphic words, Sturmian words (where the companion shadow paper locates the slowest-escape candidates), and the unstructured majority, which the no-go theorem of [10] places beyond all bounded-memory arguments. The honest value of the rung is the bridge it builds: an exact, mostly machine-verified chain from the Collatz dynamics to a single value of a classical Mahler function, with the remaining gap isolated as one well-posed question (Question 7.2) in a mature theory — and a quantified certificate (Proposition 6.1) that the question sits genuinely beyond elementary methods, for the problem's own critical reason.

Reproducibility

The formal results (`eq_of_itinerary_eq`, `divergent_itinerary_aperiodic`, `dcoef_closed`, `terras_exact_form`, the parity bijection) are in `lean/Collatz/Ladder.lean`, `Shadow.lean`, and `Parity.lean` of [10] (Lean 4 + mathlib; no axioms beyond the standard three, no `sorry`). The numerical verifications — the functional equation to degree 900, the evaluation identity to 600 bits of 2-adic precision, the exact finite-level tower identities, the block law on 200 random words, the realization digit-tail statistics, and the shadow clustering of automatic words — are `analysis/mahler_experiments.py` (experiments 1–4), deterministic, run time under a minute. The attack plan in extended form, including the failed routes, is `analysis/RUNG1_ATTACK.md`.

Acknowledgments

The mathematics, the formalization, and this text were produced in collaboration with Claude (Anthropic). The reduction owes its shape to the exact-form and cocycle infrastructure of the companion formalization; the question it isolates belongs to a theory built by Mahler and rebuilt by many hands since.

References

[1] B. Adamczewski and Y. Bugeaud, On the complexity of algebraic numbers I. Expansions in integer bases, *Ann. of Math.* **165** (2007), 547–565.

[2] B. Adamczewski and C. Faverjon, Méthode de Mahler: relations linéaires, transcendance et applications aux nombres automatiques, *Proc. Lond. Math. Soc.* **115** (2017), 55–90.

[3] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.

[4] P. Fatou, Séries trigonométriques et séries de Taylor, *Acta Math.* **30** (1906), 335–400.

[5] G. Fernandes, Méthode de Mahler en caractéristique non nulle, *Ann. Inst. Fourier* **68** (2018).

[6] J. C. Lagarias, The $3x + 1$ problem: an overview, arXiv:2111.02635 (2021).

[7] J. H. Loxton and A. J. van der Poorten, Arithmetic properties of the solutions of a class of functional equations, *J. Reine Angew. Math.* **330** (1982), 159–172.

[8] K. Mahler, Arithmetische Eigenschaften der Lösungen einer Klasse von Funktionalgleichungen, *Math. Ann.* **101** (1929), 342–366.

[9] Ku. Nishioka, *Mahler Functions and Transcendence*, Lecture Notes in Mathematics 1631, Springer, 1996.

[10] M. Sharpe, The Collatz conjecture at the critical line: machine-verified no-go, density, and cycle theorems, companion papers and repository, 2026.

[11] R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976), 241–252.